

Mini Project Objective :

The objective of this mini project is to analyze Amazon product sales and performance data using Excel and Power BI. Excel will be used for data cleaning, preprocessing, handling missing values and duplicates, standardizing data, and creating calculated fields. The cleaned data will then be used in Power BI to create interactive dashboards and identify useful insights related to products, pricing, ratings, discounts, popularity, promotions, and availability.

Dataset Information :

Particular Details

Dataset Name Amazon Product Sales Data 42K+ Items – 2025

Domain Sales & E-commerce

Source Kaggle

Number of Rows 42,675

Number of Columns 16

Raw Dataset Uncleaned dataset

Tools Used Excel & Power BI

1. Duplicate Check

The dataset was checked for duplicate records using Excel's **Remove Duplicates** feature. All 16 columns were selected for checking. **No duplicate records were found**, so no rows were removed.

2. Missing Value Analysis

Missing values were identified using the **COUNTBLANK** function. Numerical fields such as **Rating, Number of Reviews, Monthly Purchase, Discounted Price, and Original Price** were checked and handled based on the available information.

3. Rating & Number of Reviews

Missing values in **Rating** and **Number of Reviews** were handled using **median imputation**, as these fields may contain extreme values.

- Rating median = **4.5**
- Number of Reviews median = **343**

4. Monthly Purchase

The **Bought in Last Month** field contained values such as *300+ bought in past month*, *10K+ bought in past month*, etc. These were converted into numerical values in a new **Monthly Purchase** column.

For records where purchase information was unavailable, **0** was used and a separate **Purchase Data Status** column was created to distinguish available and unavailable data.

5. Price Cleaning

The price on variant field contained text and price information. The price was extracted and converted into a numerical value. Missing Discounted Price values were filled using the available variant price where appropriate. Since the **price on variant** column was not required for the final analysis and its relevant price information was already captured in the price fields, the column was removed to simplify the dataset.

6. Original Price

Currency symbols and commas were removed from **Original Price** to convert it into numerical format. Where **No Discount** was present, the applicable selling price was used as the Original Price.

7. Delivery Date

The Delivery Date column contained several "Not Available" values. Instead of creating estimated dates, the missing values were filled using the available collected_at date. A new **Cleaned Delivery Date** column was created while retaining the original Delivery Date column for reference.

8. Buy Box Availability

The Buy Box Availability column contained many "Not Available" values. Since these values represent unavailable information rather than a numerical value, they were retained as "Not Available" instead of replacing them with zero.

Pivot Tables & Charts

The cleaned dataset was analyzed using Pivot Tables and Pivot Charts in Excel to identify key patterns.

1. **Product Count by Rating** – Shows the number of products available at each rating level.
2. **Sponsored vs Organic Products** – Compares the number of sponsored and organic products.
3. **Rating vs Average Number of Reviews** – Compares product ratings with the

average number of customer reviews to understand the relationship between ratings and popularity.

Phase 2 – Power BI Data Preparation

The cleaned Excel dataset was imported into Power BI. Unnecessary blank columns were removed/hidden from the model, and the required fields were checked for correct data types

1. Calculated Columns

Discount % A calculated column was created to determine the percentage discount for each product using the original price and discounted price.

Price Category Products will be categorized into Low, Medium, and High price ranges based on their discounted price.

Rating Category Products will be categorized into **Poor, Average, Good, and Excellent** based on their rating

Review Category Products will be grouped into Low, Moderate, High, and Very High review categories based on their number of customer reviews.

2. Measures

Measures were created in Power BI to calculate key product, review, purchase, pricing, and performance metrics.

Product Measures

- Total Products
- Average Rating
- Total Reviews
- Average Reviews

Purchase Measures

- Total Monthly Purchases
- Average Monthly Purchases

Pricing Measures

- Average Original Price
- Average Discounted Price
- Average Discount %

Performance Measures

- Best Seller Products
- Sponsored Products
- Couponed Products

3. Page 1 – Executive Overview

An interactive **Executive Overview** dashboard was created to provide a quick summary of Amazon product performance.

KPI Cards

- Total Products
- Average Rating
- Total Reviews
- Average Monthly Purchases
- Average Discount %

Visualizations

- Product Count by Rating
- Sponsored vs Organic Products
- Couponed vs Non-Couponed Products
- Products by Price Category

Slicers

- Rating
- Sponsored Status
- Coupon Status
- Best Seller Status

Navigation

A page navigator was added to allow users to move between the **Executive Overview**, **Product Analysis**, **Pricing Analysis**, and **Performance Analysis** pages.

4. Page 2 – Product Analytics

An interactive **Product Analytics** dashboard was created to provide a detailed analysis of

product ratings, customer reviews, and purchasing behavior.

KPI Cards

- Average Rating
- Total Reviews
- Average Reviews per Product
- Average Monthly Purchases

Visualizations

- Product Count by Rating Category
- Average Reviews by Rating Category
- Rating vs Number of Reviews
- Average Monthly Purchases by Rating Category

Slicers

- Rating Category
- Price Category
- Sponsored Status

5. Page 3 – Pricing & Discount Analytics

An interactive **Pricing & Discount Analytics** dashboard was created to analyze product pricing, discounts, and promotional activity.

KPI Cards

- Average Original Price
- Average Discounted Price
- Average Discount %
- Maximum Discount %

Visualizations

- Average Price by Rating
- Average Discount % by Rating
- Price vs Number of Reviews
- Couponed vs Non-Couponed Products

Slicers

- Rating
- Coupon Status
- Sponsored Status
- Price Category

4. Page 4 – Performance Analysis

An interactive **Performance Analysis** dashboard was created to evaluate product demand, customer engagement, best-seller performance, and product characteristics.

KPI Cards

- Total Products
- Average Monthly Purchases
- Average Rating
- Total Reviews
- Best Seller Products

Visualizations

- Average Monthly Purchases by Rating Category
- Sponsored vs Organic Products
- Total Reviews by Rating Category
- Products by Price Category
- Best Seller vs Other Products
- Rating vs Average Monthly Purchases

Slicers

- Rating Category
- Price Category
- Sponsored Status
- Coupon Status
- Best Seller Status

5. Page 5 – Product Details

An interactive **Product Details** dashboard was created to provide detailed information about an individual product selected by the user.

KPI Cards

- Product Rating

- Total Reviews
- Monthly Purchases
- Discounted Price

Product Information

- Product Title
- Rating
- Number of Reviews
- Monthly Purchases
- Original Price
- Discounted Price

Product Status

- Selected Coupon Status
- Selected Sponsored Status
- Selected Best Seller Status

Drill-through

- **Drill-through field:** Product Title
- Users can **right-click on a product and select Drill-through** → **Product Details** to navigate to the detailed product page.
- The selected **Product Title** is passed as a filter, allowing the page to dynamically display the product's rating, reviews, monthly purchases, pricing, and promotional status.

6. Page 6 – Insights & Recommendations

An **Insights & Recommendations** page was created to summarize the key findings from the Amazon product analysis and provide actionable recommendations.

Key Insights

- **Strong Product Ratings** – The average product rating is 4.40, with most products falling into the Excellent rating category.
- **High Review Volume** – The dataset contains approximately 129M total reviews, indicating substantial product engagement.
- **Limited Coupon Usage** – Approximately 98% of products have no coupon, indicating limited use of coupon-based promotions.

- **Mostly Organic Listings** – Around 96% of products are classified as Organic, while approximately 4% are Sponsored.

Recommendations

- **Strengthen High-Rated Products** – Maintain product quality and increase visibility for highly rated products.
- **Optimize Discount Strategy** – Analyze products with higher discounts and monthly purchases to identify discount levels that can improve demand.
- **Improve Low-Rated Products** – Identify lower-rated products and focus on product quality, descriptions, and overall product experience.
- **Leverage Customer Reviews** – Use products with strong review volumes and ratings as benchmarks to understand factors associated with better product performance.